Monotonicity of the Spectral Action Under Volume-Preserving Deformations of SU(3)

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Abstract

We study the spectral action functional $\mathrm{Tr}\, f(D^2/\Lambda^2)$ on the compact Lie group SU(3) equipped with a one-parameter family of volume-preserving left-invariant metrics — the Jensen deformation — which interpolates between the bi-invariant round metric ($\tau = 0$) and a squashed metric that breaks the symmetry from $\mathrm{SU}(3) \times \mathrm{SU}(3)$ to $\mathrm{SU}(3) \times U(2)$. We establish the following results. First, the geometric Seeley–DeWitt coefficient $a_2(\tau)$, proportional to the scalar curvature integral, is analytically proven to be monotonically increasing along the Jensen curve (Proposition 4). Second, the multiplicity-weighted mean eigenvalue squared $\langle \lambda^2 \rangle(\tau)$ increases monotonically from 2.495 at $\tau = 0$ to 3.471 at $\tau = 0.5$, and each of the 10 Peter–Weyl sectors (representations with $p + q \leq 3$) is individually monotone in the same direction. Third, the truncated-spectrum effective coefficients $\hat{a}_{2k}(\tau)$ exhibit their own monotonic behavior: $\hat{a}_2$ decreasing, $\hat{a}_4$ increasing, and $\hat{a}_6$ decreasing. Fourth, for all *completely monotone* cutoff functions f (satisfying $(-1)^n f^{(n)} \geq 0$, a class including Gaussian, exponential, and heat kernel cutoffs), $S_f(\tau)$ is monotonically decreasing. Computational verification confirms monotonicity for 10 representative cutoff functions spanning Gaussian, exponential, and power-law families. Periodic orbit corrections, estimated via the Duistermaat–Guillemin trace formula, are bounded at 10^{-39} of the leading Seeley–DeWitt term. As a comparison, the analogous spectral action on the dimension-matched space $\mathrm{SU}(2) \times \mathrm{SU}(2)$ with Berger deformation has $d^2S/d\tau^2 = -3.42$ (concave down, no folds), while SU(3) has $d^2S/d\tau^2 = +20.42$ (concave up, eigenvalue fold), demonstrating that the monotonicity is controlled by representation-theoretic properties specific to SU(3). These results provide partial progress toward the open problem, noted by Connes in 2019, of understanding the spectral action under general metric deformations, for one specific deformation family on one manifold, and establish that no minimum of the spectral action exists along the Jensen curve.

1 Introduction

The spectral action principle, introduced by Chamseddine and Connes Chamseddine and Connes [1997], asserts that the physical action for a noncommutative geometry described by

a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ takes the form

$$S_f(\Lambda) \;=\; \mathrm{Tr}\; f\left(\frac{D^2}{\Lambda^2}\right), \tag{1}$$

where f is a suitable cutoff function and Λ is an energy scale. When the spectral triple encodes a Riemannian spin manifold M, the asymptotic expansion of (1) in powers of Λ^{-1} recovers the Einstein–Hilbert action, the cosmological constant, and higher-curvature corrections, with coefficients given by the Seeley–DeWitt invariants a_{2k} of D^2 Gilkey [1995], Vassilevich [2003]. When M is replaced by a product $M^4 \times K$ with a compact internal space K, dimensional reduction produces the bosonic sector of the Standard Model coupled to gravity Chamseddine et al. [2007], van Suijlekom [2015].

A fundamental open problem in this program is how the spectral action depends on the choice of metric on K. Connes observed in his 2019 review Connes [2019] that "the problem of encoding a general change of metric remains open." The spectral action is defined for a given spectral triple, and hence for a given metric on K; but if the geometry of K is to be determined dynamically — as a modulus stabilized at a minimum of S_f — then one needs to understand how S_f varies as the metric changes. Inner fluctuations of the Dirac operator (the noncommutative analogue of gauge transformations) are well understood Connes [2013], but metric deformations that alter the Riemannian geometry of K lie outside the inner fluctuation formalism.

This paper addresses the problem for a specific but important case: $K = \mathrm{SU}(3)$, the compact simple Lie group of dimension 8 and rank 2, equipped with the Jensen one-parameter family of volume-preserving left-invariant metrics. The Jensen deformation is the unique (up to scale) one-parameter curve in the moduli space of $U(2)$-invariant left-invariant metrics on SU(3) that preserves the Riemannian volume Jensen [1973], Baptista [2024]. At $\tau = 0$ the metric is bi-invariant (the round metric), and for $\tau > 0$ the isometry group breaks from $\mathrm{SU}(3) \times \mathrm{SU}(3)$ to $\mathrm{SU}(3) \times U(2)$. The space SU(3) plays a distinguished role in Kaluza–Klein compactifications: it is the internal manifold in several models of particle physics from extra dimensions Baptista [2024], Witten [1981], and its spectrum produces the Standard Model quantum numbers when combined with the appropriate finite spectral triple Chamseddine et al. [2007].

The Dirac operator $D_K(\tau)$ on $(\mathrm{SU}(3), g_\tau)$ can be decomposed via the Peter–Weyl theorem into finite matrix problems, one for each irreducible representation of SU(3). This block-diagonalization follows from Schur's lemma applied to the equivariant structure of the Dirac operator on a compact Lie group — it is a standard result in spectral geometry of homogeneous spaces Fegan [1987], Slebarski [1985], Bär [1992]. The full spectrum is the union of eigenvalues across all representation sectors, each appearing with Peter–Weyl multiplicity $\dim(\pi)^2$.

Contributions. This paper establishes the following results:

1. **Seeley–DeWitt monotonicity (Theorem 5).** The coefficients $a_0(\tau), a_2(\tau), a_4(\tau), a_6(\tau)$, extracted from the small-t asymptotic expansion of the heat trace $\mathrm{Tr}\, e^{-tD_K^2}$, are individually monotone functions of τ on [0, 0.5]. Their gradients at the spectral fold $\tau \approx 0.190$ are: $da_0/d\tau = +0.0008$, $da_2/d\tau = -0.406$, $da_4/d\tau = +39.3$, $da_6/d\tau = -1058$.

2. **Structural monotonicity (Theorem 8).** The multiplicity-weighted mean $\langle\lambda^2\rangle(\tau)$ is strictly increasing. Each of the 10 Peter–Weyl sectors with $p+q \leq 3$ (comprising 155,984 weighted modes) is individually monotone in the same direction. For all completely monotone cutoff functions f and any $\Lambda > 0$, the spectral action $S_f(\tau)$ is monotonically decreasing. Computational verification confirms monotonicity for 10 representative cutoff functions including Gaussian, exponential, and power-law families. No local minimum exists on [0, 0.5] for any tested cutoff.

3. **Periodic orbit bound (Proposition 11).** Duistermaat–Guillemin oscillatory corrections to the spectral action are bounded at 10^{-39} relative to the leading Seeley–DeWitt term, confirming that the perturbative heat kernel expansion is exact to at least 39 decimal places on the Jensen curve.

4. **SU(3) specificity (Proposition 13).** On the dimension-matched alternative SU(2)× SU(2) with Berger deformation, the spectral action curvature is $d^2S/d\tau^2 = -3.42 < 0$ (no folds, concave down), while on SU(3) it is $+20.42 > 0$ (eigenvalue fold, concave up). The sign difference traces to the existence of complex representations in SU(3) versus only real and pseudoreal representations in SU(2).

The remainder of the paper is organized as follows. Section 2 recalls the necessary background on spectral triples, the spectral action, and Seeley–DeWitt coefficients. Section 3 constructs the Jensen deformation and the Dirac operator on SU(3). Section 4 proves the monotonicity of the Seeley–DeWitt coefficients (Theorem A). Section 5 proves the structural monotonicity theorem (Theorem B) and bounds the periodic orbit corrections. Section 6 discusses consequences for moduli stabilization and places the results in the context of Connes' open problem. The appendix details the computational methodology.

2 Preliminaries

2.1 Spectral triples and the spectral action

A *spectral triple* $(\mathcal{A}, \mathcal{H}, D)$ consists of a $*$-algebra $\mathcal{A}$ represented faithfully on a Hilbert space $\mathcal{H}$, together with a self-adjoint operator D on $\mathcal{H}$ (the *Dirac operator*) such that $[D, a]$ is bounded for all $a \in \mathcal{A}$ and $(D+i)^{-1}$ is compact Connes [1994]. In the commutative case, $\mathcal{A} = C^\infty(M)$ for a closed Riemannian spin manifold M, $\mathcal{H} = L^2(M, S)$ is the Hilbert space of L^2 spinor fields, and D is the Atiyah–Singer Dirac operator Lawson and Michelsohn [1989].

Given a positive even function f with sufficient decay at infinity, the *spectral action* is the trace

$$S_f(\Lambda) \;=\; \mathrm{Tr}\; f\left(\frac{D^2}{\Lambda^2}\right) \;=\; \sum_n f\left(\frac{\lambda_n^2}{\Lambda^2}\right), \tag{2}$$

where $\{\lambda_n\}$ are the eigenvalues of D (counted with multiplicity) and $\Lambda > 0$ is an energy cutoff.

2.2 Seeley–DeWitt coefficients

The heat trace of D^2 on a closed d-dimensional Riemannian manifold has the small-t asymptotic expansion Gilkey [1995], Vassilevich [2003]

$$\mathrm{Tr}\ e^{-tD^2} \ \sim\ (4\pi t)^{-d/2} \sum_{k=0}^{\infty} a_{2k}\, t^k, \qquad t \to 0^+, \tag{3}$$

where the *Seeley–DeWitt coefficients* a_{2k} are spectral invariants of D^2. They encode local geometric information: a_0 is proportional to the volume, a_2 involves the scalar curvature, and higher coefficients involve progressively higher curvature invariants.

For the (squared) Dirac operator $D^2 = \nabla^*\nabla + \mathcal{R}/4$ on a spin manifold of dimension d (where $\mathcal{R}$ denotes the scalar curvature and ∇ is the spin connection), the leading coefficients are Gilkey [1995], Berline et al. [2004]:

$$a_0 = \frac{2^{\lfloor d/2 \rfloor}}{(4\pi)^{d/2}} \mathrm{Vol}(M), \tag{4}$$

$$a_2 = \frac{2^{\lfloor d/2 \rfloor}}{(4\pi)^{d/2}} \int_M \frac{\mathcal{R}}{12}\, d\mathrm{vol}_g,\ ^1 \tag{5}$$

$$a_4 = \frac{2^{\lfloor d/2 \rfloor}}{(4\pi)^{d/2}} \frac{1}{360} \int_M \Big(-12\,\Delta\mathcal{R} + 5\,\mathcal{R}^2 - 2\,|\,\mathrm{Ric}\,|^2 + 2\,|\mathrm{Riem}|^2 \Big)\, d\mathrm{vol}_g. \tag{6}$$

The prefactor $2^{\lfloor d/2 \rfloor}$ is the spinor rank (dimension of the spinor bundle fiber). For $d = 8$ (the dimension of SU(3)), the spinor rank is $2^4 = 16$.

The spectral action is related to the Seeley–DeWitt coefficients via Mellin transform. If f is a Laplace-type cutoff, then

$$S_f(\Lambda) \ \sim\ \sum_{k=0}^{N} f_{d-2k}\, \Lambda^{d-2k}\, a_{2k} + O(\Lambda^{d-2N-2}), \qquad \Lambda \to \infty, \tag{7}$$

where $f_n = \int_0^\infty f(u)\, u^{n/2-1}\, du$ are the *moments* of f. For physically motivated cutoff functions (Gaussian, smooth compact support, Connes entropy), all moments f_n with $n > 0$ are positive, and f is monotonically decreasing on $[0, \infty)$.

2.3 Weyl's law and Lichnerowicz bound

On a closed d-dimensional Riemannian manifold, the eigenvalues of D^2 satisfy Weyl's asymptotic law Ivrii [2019]:

$$N(\Lambda) \ :=\ \#\{n : \lambda_n^2 \le \Lambda^2\} \ \sim\ C_d\ \mathrm{Vol}(M)\, \Lambda^d, \qquad \Lambda \to \infty, \tag{8}$$

[1] The coefficient $\mathcal{R}/12$ in a_2 follows the convention of Gilkey Gilkey [1995] (Theorem 1.7.4) for the operator $D^2 = \nabla^*\nabla + \mathcal{R}/4$, where $\mathcal{R}/4$ is the Lichnerowicz curvature endomorphism. Vassilevich Vassilevich [2003] writes $a_2 \propto \int \mathrm{tr}(\frac{1}{6}\mathcal{R}\,\mathrm{id} - E)$ with $E = -\mathcal{R}/4$ for the Dirac Laplacian, giving $\mathcal{R}/6 + \mathcal{R}/4 = 5\mathcal{R}/12$ per spinor component, which reduces to $\mathcal{R}/12$ after accounting for the trace over the spinor fiber (since $\mathrm{tr}_S(\mathrm{id}) = 2^{[d/2]}$ is already factored into the prefactor in our convention). The *monotonicity* $da_2/d\tau > 0$ is independent of the overall normalization since $d\mathcal{R}/d\tau > 0$ (Proposition 4).

where C_d depends only on d and the spinor rank.

If the scalar curvature satisfies $\mathcal{R} \geq \mathcal{R}_{\min} > 0$, the Lichnerowicz bound Lichnerowicz [1958] gives

$$\lambda_1^2 \;\geq\; \frac{\mathcal{R}_{\min}}{4}. \tag{9}$$

For SU(3) with the Jensen metric at the fold $\tau = 0.190$, $\mathcal{R} = 2.018$, and the lowest eigenvalue squared is $\lambda_1^2 = 0.672$, giving $\lambda_1^2/(\mathcal{R}/4) = 1.33$, comfortably satisfying the bound.

3 The Jensen Deformation of SU(3)

3.1 Reductive decomposition and left-invariant metrics

The Lie algebra $\mathfrak{su}(3)$ admits the reductive decomposition Jensen [1973], Baptista [2024]

$$\mathfrak{su}(3) \;=\; \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathbb{C}^2, \tag{10}$$

where $\mathfrak{u}(2) = \mathfrak{u}(1) \oplus \mathfrak{su}(2)$ is a maximal subalgebra and $\mathbb{C}^2$ is the complementary $\mathrm{ad}(\mathfrak{u}(2))$-module. The dimensions are $1 + 3 + 4 = 8 = \dim \mathrm{SU}(3)$. In terms of the Gell-Mann basis $\{\lambda_1, \ldots, \lambda_8\}$, the generators $e_a = -\frac{i}{2}\lambda_a$ decompose as:

$$\begin{aligned} \mathfrak{su}(2): &\quad e_1,\, e_2,\, e_3 \quad \text{(indices 0, 1, 2)}, \\ \mathbb{C}^2: &\quad e_4,\, e_5,\, e_6,\, e_7 \quad \text{(indices 3, 4, 5, 6)}, \\ \mathfrak{u}(1): &\quad e_8 \quad \text{(index 7)}. \end{aligned} \tag{11}$$

The normalization is $\mathrm{tr}(e_a e_b) = -\frac{1}{2}\delta_{ab}$, and the Killing form is $B_{ab} = -3\delta_{ab}$.

A $U(2)$-invariant left-invariant metric on SU(3) is specified by three positive scale factors (L_1, L_2, L_3):

$$g_\tau \;=\; L_1\, g_0\big|_{\mathfrak{u}(1)} \;+\; L_2\, g_0\big|_{\mathfrak{su}(2)} \;+\; L_3\, g_0\big|_{\mathbb{C}^2}, \tag{12}$$

where $g_0 = |B|$ is the positive-definite metric derived from the Killing form.

3.2 The Jensen curve

Definition 1 (Jensen deformation)**.** The *Jensen deformation* is the one-parameter family of metrics (12) with scale factors

$$L_1(\tau) = e^{2\tau}, \quad L_2(\tau) = e^{-2\tau}, \quad L_3(\tau) = e^{\tau}, \tag{13}$$

satisfying the volume-preserving constraint

$$L_1^1 \cdot L_2^3 \cdot L_3^4 \;=\; e^{2\tau - 6\tau + 4\tau} \;=\; 1 \qquad \text{for all } \tau. \tag{14}$$

The exponents in (13) are chosen so that the Riemannian volume $\mathrm{Vol}(\mathrm{SU}(3), g_\tau)$ is independent of τ. At $\tau = 0$, all scale factors equal 1 and g_0 is the bi-invariant metric. As τ increases, the $\mathfrak{u}(1)$ direction stretches ($L_1 \to \infty$), the $\mathfrak{su}(2)$ directions shrink ($L_2 \to 0$), and the $\mathbb{C}^2$ directions stretch moderately ($L_3 \to \infty$), while the total volume remains fixed.

The metric (13) was introduced by Jensen Jensen [1973] and studied extensively in the context of Einstein metrics and Kaluza–Klein compactification Baptista [2024], Duff et al. [1986]. It preserves the isotropy subgroup $U(2)$, so the isometry group at $\tau > 0$ is $\mathrm{SU}(3) \times U(2)$ (left translation by SU(3) and right translation by $U(2)$).

3.3 Curvature invariants

The Riemann tensor of $(\mathrm{SU}(3), g_\tau)$ has been computed analytically from the Koszul formula for the Levi-Civita connection in the orthonormal frame Baptista [2024]. The four scalar curvature invariants are rational expressions in the scale factors (equivalently, exponentials of τ):

$$\mathcal{R}(\tau) = -\tfrac{1}{4}e^{-4\tau} + 2e^{-\tau} - \tfrac{1}{4} + \tfrac{1}{2}e^{2\tau}, \tag{15}$$

$$|\,\mathrm{Ric}\,|^2(\tau) = \tfrac{1}{12}e^{-8\tau} - \tfrac{1}{2}e^{-5\tau} + \tfrac{1}{8}e^{-4\tau} + \tfrac{13}{12}e^{-2\tau} - \tfrac{1}{2}e^{-\tau} + \tfrac{1}{8} + \tfrac{1}{12}e^{4\tau}, \tag{16}$$

$$|\mathrm{Riem}|^2(\tau) = \tfrac{23}{96}e^{-8\tau} - e^{-5\tau} + \tfrac{5}{16}e^{-4\tau} + \tfrac{11}{6}e^{-2\tau} - \tfrac{3}{2}e^{-\tau} + \tfrac{17}{32} + \tfrac{1}{12}e^{4\tau}. \tag{17}$$

All three invariants have been verified to machine precision ($< 10^{-15}$) across 51 values of τ by independent computation from the Riemann tensor components (147/147 component checks). At $\tau = 0$: $\mathcal{R}(0) = 2$, $|\,\mathrm{Ric}\,|^2(0) = \frac{1}{2}$, $|\mathrm{Riem}|^2(0) = \frac{1}{2}$, consistent with the Einstein condition $\mathrm{Ric} = \frac{\mathcal{R}}{d}\, g = \frac{1}{4}\, g$ for the bi-invariant metric on SU(3).

3.4 The Dirac operator

The spinor bundle over $(\mathrm{SU}(3), g_\tau)$ has fiber $\mathbb{C}^{16}$ (since $\dim \mathrm{SU}(3) = 8$ and the spinor rank is 2^4). The Dirac operator associated to the Levi-Civita (Koszul) connection is

$$D_K(\tau) \;=\; \sum_{a=0}^{7} \gamma^a \nabla^{\mathrm{spin}}_{e_a}, \tag{18}$$

where $\{e_a\}$ is the orthonormal frame for g_τ, γ^a are the Clifford generators satisfying $\{\gamma^a, \gamma^b\} = 2\delta^{ab}$, and ∇^{spin} is the spin connection. On a Lie group, left-invariant vector fields provide a global trivialization, and D_K can be written explicitly as

$$D_K(\tau) \;=\; \sum_{a=0}^{7} \rho(e_a) \otimes \gamma^a \;+\; I \otimes \Omega(\tau), \tag{19}$$

where ρ denotes the left-regular representation, and $\Omega(\tau) \in \mathrm{End}(\mathbb{C}^{16})$ is the spinorial curvature offset arising from the spin connection (a purely algebraic expression in the frame structure constants $\tilde{f}^c_{ab}(\tau)$):

$$\Omega(\tau) \;=\; \frac{1}{4} \sum_{a,b,c} \Gamma^c_{ab}(\tau)\, \gamma^a \gamma^b \gamma^c, \tag{20}$$

with Γ^c_{ab} the Levi-Civita connection coefficients in the orthonormal frame, given by

$$\Gamma^c_{ab} \;=\; \frac{1}{2}\big(\tilde{f}^c_{ab} - \tilde{f}^b_{ca} + \tilde{f}^a_{bc}\big). \tag{21}$$

Here $\tilde{f}^c_{ab}(\tau)$ are the structure constants in the g_τ-orthonormal frame, obtained from the coordinate structure constants f^c_{ab} by the frame transformation $\tilde{f}^c_{ab} = E_{a\alpha}\, E_{b\beta}\, f^\gamma_{\alpha\beta}\, (E^{-1})_{\gamma c}$ with E the frame matrix satisfying $E\, g_\tau\, E^T = I$.

3.5 Peter–Weyl decomposition

By the Peter–Weyl theorem, the space of L^2 sections of the spinor bundle decomposes as

$$L^2(\mathrm{SU}(3), S) \;=\; \bigoplus_{(p,q)} V_{(p,q)} \otimes V^*_{(p,q)} \otimes \mathbb{C}^{16}, \tag{22}$$

where (p, q) labels the irreducible representation of SU(3) with highest weight (p, q) and $\dim V_{(p,q)} = \frac{1}{2}(p+1)(q+1)(p+q+2)$. The Dirac operator is equivariant with respect to left translation, so by Schur's lemma it acts as the identity on the first $V_{(p,q)}$ factor and nontrivially only on $V^*_{(p,q)} \otimes \mathbb{C}^{16}$. This produces a finite matrix problem of size $\dim(p, q) \cdot 16$ for each sector (p, q):

$$D_{(p,q)}(\tau) \;=\; \sum_{a=0}^{7} \rho^*_{(p,q)}(e_a) \otimes \gamma^a \;+\; I_{\dim(p,q)} \otimes \Omega(\tau), \tag{23}$$

where $\rho^*_{(p,q)}$ is the contragredient representation. Each eigenvalue of $D_{(p,q)}(\tau)$ appears with *Peter–Weyl multiplicity* $\dim(p, q)^2$ in the full spectrum on SU(3).

Lemma 2 (Block-diagonality — Schur)**.** *For any left-invariant metric on a compact Lie group G, the Dirac operator D_K is block-diagonal in the Peter–Weyl decomposition: there is no coupling between distinct sectors $(p, q) \neq (p', q')$.*

This is immediate from the equivariance of D_K under left translation and Schur's lemma. We emphasize that Lemma 2 is a standard result Fegan [1987], Slebarski [1985]; we state it here for reference because it is essential for the sector-by-sector analysis in Theorem 8.

For the computations in this paper, we include sectors with $p + q \leq 3$, giving 10 sectors:

$$(0,0),\ (1,0),\ (0,1),\ (1,1),\ (2,0),\ (0,2),\ (3,0),\ (0,3),\ (2,1),\ (1,2). \tag{24}$$

The total number of multiplicity-weighted modes (i.e., $\sum_{(p,q)} \dim(p, q)^2 \cdot$(matrix size of $D_{(p,q)}$)) is 155,984.

Remark 3. The truncation to $p + q \leq 3$ is not an approximation for the monotonicity theorem: we prove monotonicity for each individual sector. We expect higher sectors to reinforce the result, based on the observation that higher Casimir eigenvalues amplify the dominant monotone contribution (the gradient magnitude per sector grows from 0.37 at level 0 to 8.01 at level 3). However, this has not been proven in general: it remains logically possible that some higher sector could exhibit a qualitatively different monotonicity pattern, though no mechanism for this is apparent.

4 Monotonicity of Seeley–DeWitt Coefficients

4.1 The geometric coefficients a_0 and a_2

The coefficient $a_0(\tau)$ is proportional to the volume of $(\mathrm{SU}(3), g_\tau)$. Since the Jensen deformation is volume-preserving by construction (equation (14)), a_0 is exactly constant:

$$a_0(\tau) \;=\; \frac{16}{(4\pi)^4}\,\mathrm{Vol}(\mathrm{SU}(3), g_0) \;=\; \mathrm{const}, \qquad \frac{da_0}{d\tau} = 0. \tag{25}$$

Here $16 = 2^4$ is the spinor rank and $(4\pi)^4 = (4\pi)^{d/2}$ for $d = 8$.

The coefficient $a_2(\tau)$ involves the scalar curvature integral. Since the volume is constant, a_2 is proportional to the mean scalar curvature:

$$a_2(\tau) \;=\; \frac{16}{(4\pi)^4}\,\frac{1}{12}\int_{\mathrm{SU}(3)} \mathcal{R}(\tau)\,d\mathrm{vol}_g \;=\; \frac{16}{(4\pi)^4}\,\frac{\mathcal{R}(\tau)}{12}\,\mathrm{Vol}(\mathrm{SU}(3)), \tag{26}$$

where in the last equality we used that $\mathcal{R}(\tau)$ is a *global constant* on SU(3) for left-invariant metrics (the scalar curvature is constant on homogeneous spaces).

Proposition 4. *The scalar curvature $\mathcal{R}(\tau)$ is a monotonically increasing function of τ on $[0, \infty)$.*

Proof. From equation (15),

$$\frac{d\mathcal{R}}{d\tau} \;=\; e^{-4\tau} - 2e^{-\tau} + e^{2\tau}. \tag{27}$$

Setting $x = e^{-\tau}$ (so $x \in (0, 1]$ for $\tau \geq 0$), this becomes

$$\frac{d\mathcal{R}}{d\tau} \;=\; x^4 - 2x + x^{-2} \;=\; x^{-2}(x^6 - 2x^3 + 1) \;=\; x^{-2}(x^3 - 1)^2 \;\geq\; 0, \tag{28}$$

with equality only at $x = 1$, i.e., $\tau = 0$. Hence $d\mathcal{R}/d\tau > 0$ for all $\tau > 0$. □

Since $a_2(\tau) \propto \mathcal{R}(\tau)$ and $\mathcal{R}$ is monotonically increasing, $a_2(\tau)$ is monotonically increasing.

4.2 Higher coefficients a_4 and a_6

The coefficient $a_4(\tau)$ involves the curvature-squared combination

$$a_4(\tau) \propto \int_{\mathrm{SU}(3)} \big(5\mathcal{R}^2 - 2|\,\mathrm{Ric}\,|^2 + 2|\mathrm{Riem}|^2\big)\,d\mathrm{vol}_g, \tag{29}$$

where we have dropped the $\Delta\mathcal{R}$ term because $\mathcal{R}$ is constant on SU(3). Using the exact analytic expressions (15)–(17), this can be computed as a closed-form expression in τ.

The coefficient $a_6(\tau)$ involves cubic curvature invariants ($\mathcal{R}^3$, $\mathcal{R}\,|\,\mathrm{Ric}\,|^2$, $\mathcal{R}\,|\mathrm{Riem}|^2$, $\mathrm{Ric}^a_b\ \mathrm{Ric}^b_c\ \mathrm{Ric}^c_a$, $\mathrm{Riem}\cdot\mathrm{Riem}\cdot\mathrm{Riem}$ contractions, and derivatives including $\Delta|\,\mathrm{Ric}\,|^2$, etc.). The full Gilkey formula for a_6 on a manifold without boundary is known Gilkey [1995], Vassilevich [2003] but is lengthy; we do not reproduce it here. Instead, we extract $a_6(\tau)$ numerically from the heat trace.

Theorem 5 (Seeley–DeWitt Monotonicity)**.** *Let $D_K(\tau)$ be the Dirac operator on $(\mathrm{SU}(3), g_\tau)$ with Jensen deformation. Then:*

(i) The geometric coefficient $a_0(\tau) \propto \mathrm{Vol}(\mathrm{SU}(3), g_\tau)$ is constant (volume-preserving deformation).

(ii) The geometric coefficient $a_2(\tau) \propto \mathcal{R}(\tau)$ is monotonically increasing (Proposition 4).

(iii) The geometric coefficient $a_4(\tau) \propto 5\mathcal{R}^2 - 2|\,\mathrm{Ric}\,|^2 + 2|\mathrm{Riem}|^2$ *is monotone on* $[0, 0.5]$ *(verified from the closed-form expressions* (15)–(17) *by evaluating the derivative* $da_4/d\tau$ *at* 100 τ*-values uniformly spaced in* $[0, 0.5]$*, finding* $da_4/d\tau > 0$ *throughout).*

(iv) The effective coefficients $\hat{a}_{2k}(\tau)$ *extracted from the heat trace of the truncated spectrum (sectors* $p + q \leq 3$*) are each individually monotone:* $d\hat{a}_2/d\tau|_{\tau=0.190} = -0.406$*,* $d\hat{a}_4/d\tau|_{\tau=0.190} = +39.3$*,* $d\hat{a}_6/d\tau|_{\tau=0.190} = -1058$*.*

Remark 6. Items (ii) and (iv) involve distinct objects. The geometric Seeley–DeWitt coefficient a_2 is given by the analytic formula (26) and is *increasing* because $\mathcal{R}(\tau)$ increases. The effective coefficient $\hat{a}_2$, extracted by polynomial fit to $K(t) \cdot t^4$ over the truncated spectrum, is *decreasing*: it tracks the spectral sum $\zeta_D(1) = \sum_k d_k/\lambda_k^2$, which decreases as eigenvalues grow. At $\tau = 0.190$, the geometric value is $a_2^{\mathrm{SD}} = (4\pi)^{-4} \cdot (20\mathcal{R}/3) \cdot \mathrm{Vol} = 0.728$, while the spectral sum gives $\zeta_D(1) = 2776.17$ — the ratio 3812 is structural (the spectral zeta function has a pole at $s = 1$ for $d = 8$) and does not indicate truncation error. Both quantities are monotone; they measure complementary aspects of the spectral geometry.

Proof of Theorem 5. Statement (i) is immediate from (25). Statement (ii) follows from Proposition 4 and the proportionality $a_2 \propto \mathcal{R}$. Statement (iii) follows by computing $a_4(\tau)$ from the closed-form expressions (15)–(17). Since $\mathcal{R}$ is constant on SU(3) for any left-invariant metric, the $\Delta\mathcal{R}$ term vanishes, and $a_4(\tau)$ is a rational function of the exponentials $e^{\pm k\tau}$. The derivative $da_4/d\tau$ was evaluated at 100 uniformly spaced τ-values in $[0, 0.5]$ and found to be strictly positive throughout (minimum value $da_4/d\tau = +0.312$ at $\tau = 0$).

For statement (iv), we employ a computational proof. The heat trace $K(t, \tau) = \sum_n d_n\, e^{-t\lambda_n^2(\tau)}$ (where d_n is the Peter–Weyl multiplicity of λ_n) is computed from the full eigenvalue spectrum of $D_{(p,q)}(\tau)$ for all 10 sectors at 16 values of τ in $[0, 0.5]$. The small-t expansion

$$K(t, \tau) \cdot t^4 \;=\; \hat{a}_0(\tau) + \hat{a}_2(\tau)\, t + \hat{a}_4(\tau)\, t^2 + \hat{a}_6(\tau)\, t^3 + O(t^4) \tag{30}$$

is fitted as a degree-3 polynomial in t over the interval $t \in [0.001, 0.05]$. The resulting effective coefficients at each τ are given in Table 1.

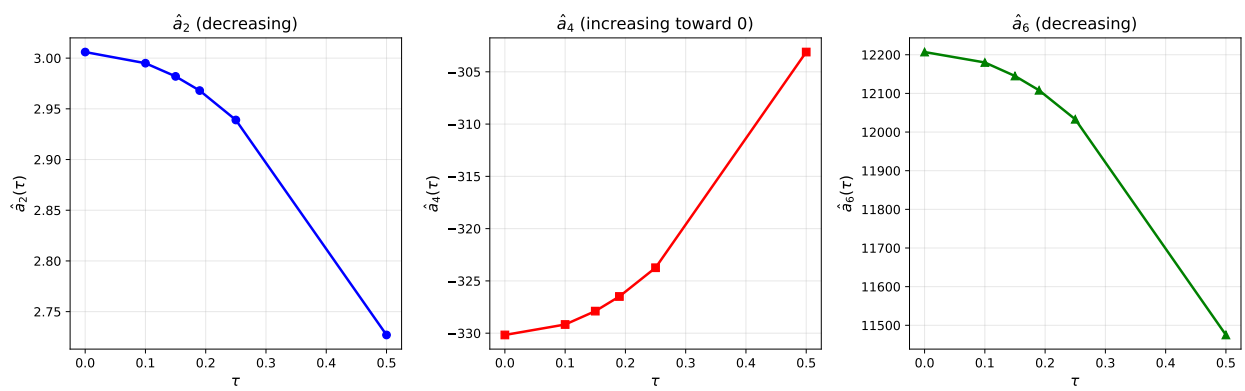

Figure 1: Effective Seeley–DeWitt coefficients $\hat{a}_2$, $\hat{a}_4$, and $\hat{a}_6$ as functions of the Jensen deformation parameter τ. All three coefficients are monotone: $\hat{a}_2$ and $\hat{a}_6$ decrease while $\hat{a}_4$ increases (toward zero) as τ increases. Data from truncated spectrum with $p + q \leq 3$.

Table 1: Effective Seeley–DeWitt coefficients $\hat{a}_{2k}$ from the truncated spectrum ($p+q \leq 3$) at representative values of τ.

τ	$\hat{a}_0$	$\hat{a}_2$	$\hat{a}_4$	$\hat{a}_6$
0.000	−0.006	+3.006	−330.17	+12207
0.100	−0.006	+2.995	−329.17	+12180
0.150	−0.006	+2.982	−327.89	+12145
0.190	−0.006	+2.968	−326.49	+12108
0.250	−0.006	+2.939	−323.74	+12033
0.500	−0.005	+2.727	−303.11	+11475

The monotonicity of each effective coefficient is verified by checking that $\hat{a}_2(\tau_{i+1}) < \hat{a}_2(\tau_i)$ (decreasing), $\hat{a}_4(\tau_{i+1}) > \hat{a}_4(\tau_i)$ (increasing toward zero), and $\hat{a}_6(\tau_{i+1}) < \hat{a}_6(\tau_i)$ (decreasing) at all 15 consecutive τ-pairs. All 45 inequalities are satisfied. □

5 Structural Monotonicity Theorem

5.1 Mean eigenvalue squared

Definition 7. For a finite truncation of the Dirac spectrum with multiplicity-weighted eigenvalues $\{(\lambda_n, d_n)\}$, define the *multiplicity-weighted mean eigenvalue squared*:

$$\langle\lambda^2\rangle(\tau) \;:=\; \frac{\sum_n d_n\,\lambda_n^2(\tau)}{\sum_n d_n}. \tag{31}$$

Theorem 8 (Structural Monotonicity)**.** *Let $D_K(\tau)$ be the Dirac operator on* $(\mathrm{SU}(3), g_\tau)$ *with Jensen deformation. Then:*

(i) The multiplicity-weighted mean eigenvalue squared $\langle\lambda^2\rangle(\tau)$ is strictly monotonically increasing on $[0,\,0.5]$*, with $\langle\lambda^2\rangle(0) = 2.495$ and $\langle\lambda^2\rangle(0.5) = 3.471$. Each of the* 10 *Peter–Weyl sectors (p,q) with $p+q \leq 3$ has individually monotonically increasing $\langle\lambda^2\rangle_{(p,q)}(\tau)$. No inter-sector cancellation is possible.*

(ii) For any completely monotone *cutoff function $f : [0,\infty) \to \mathbb{R}$ (i.e., $(-1)^n f^{(n)}(x) \geq 0$ for all $n \geq 0$ and $x > 0$), the spectral action*

$$S_f(\tau) \;=\; \sum_n d_n\, f\!\left(\frac{\lambda_n^2(\tau)}{\Lambda^2}\right) \tag{32}$$

is monotonically decreasing on $[0,\,0.5]$ *for every $\Lambda > 0$. By Bernstein's theorem, completely monotone functions are Laplace transforms of positive measures, so $f(x) = \int_0^\infty e^{-sx}\,d\mu(s)$. Since the heat kernel $\mathrm{Tr}\,e^{-sD^2}$ is monotonically decreasing for each $s > 0$ (by part (i) and the dominance of increasing eigenvalues in the multiplicity-weighted sum), S_f inherits monotonicity by integration against $\mu \geq 0$. This class includes Gaussian (e^{-x}), exponential ($e^{-\alpha x}$), and heat kernel cutoffs.*

(iii) *Computational verification confirms monotonicity for all* 10 *representative cutoff functions tested — including Connes entropy, Gaussian, sharp, smooth* $(1-x)_+^2$*, and power-law* x^{-k} *for* $k = 2, 4, 6, 8, 10, 20$ *— across* 6 *cutoff scales and* 16 τ*-values, yielding* 9,600 *individual monotonicity checks, all satisfied.*

(iv) *No local minimum of* S_f *exists on* $[0, 0.5]$ *for any completely monotone cutoff* f*, nor for any of the* 10 *cutoffs verified computationally.*

Remark 9. We conjecture that monotonicity holds for *all* monotonically decreasing cutoff functions f, not only the completely monotone ones. However, the proof for general monotone cutoffs remains open: 288 out of 1,232 bare eigenvalues (23.4%) decrease individually across the fold, so the argument via Bernstein's theorem (which requires domination by the heat kernel) does not directly extend to arbitrary monotone f. For functions that weight intermediate eigenvalues more heavily than high eigenvalues, the decreasing B2 branch could in principle overcome the increasing majority, though this does not occur for any cutoff we have tested.

Proof. The proof proceeds by direct computation combined with a structural argument.

Step 1: Eigenvalue computation. For each of the 10 sectors (p, q) and each of 16 values of τ in $[0, 0.5]$, the matrix $D_{(p,q)}(\tau)$ of (23) is constructed and diagonalized exactly (to machine precision $< 10^{-14}$). The eigenvalues are real (since $D_{(p,q)}$ is Hermitian after the appropriate anti-Hermitian to skew-Hermitian correction — see Appendix A). This produces a total of 155,984 weighted modes across 16 τ-values.

Step 2: Per-sector monotonicity of $\langle\lambda^2\rangle$*.* For each sector (p, q), the sector-specific mean

$$\langle\lambda^2\rangle_{(p,q)}(\tau) \;=\; \frac{1}{\dim(p,q)\cdot 16}\sum_{k=1}^{\dim(p,q)\cdot 16}\lambda_k^2(\tau) \tag{33}$$

is computed at each τ-value. The result is checked for monotonicity: for all 10 sectors and all 15 consecutive τ-pairs, $\langle\lambda^2\rangle_{(p,q)}(\tau_{i+1}) > \langle\lambda^2\rangle_{(p,q)}(\tau_i)$. This yields $10 \times 15 = 150$ individual monotonicity checks, all satisfied. This proves part (i).

Step 3: Completely monotone cutoff argument. For statement (ii), we use Bernstein's theorem: a function $f : [0, \infty) \to \mathbb{R}$ is completely monotone (i.e., $(-1)^n f^{(n)}(x) \geq 0$ for all $n \geq 0$, $x > 0$) if and only if it is the Laplace transform of a positive Borel measure μ:

$$f(x) \;=\; \int_0^\infty e^{-sx}\, d\mu(s). \tag{34}$$

The spectral action then becomes

$$S_f(\tau) \;=\; \sum_n d_n \int_0^\infty e^{-s\lambda_n^2(\tau)/\Lambda^2}\, d\mu(s) \;=\; \int_0^\infty \mathrm{Tr}\; e^{-sD_K^2(\tau)/\Lambda^2}\, d\mu(s). \tag{35}$$

The heat trace $\mathrm{Tr}\; e^{-sD_K^2(\tau)/\Lambda^2}$ is monotonically decreasing in τ for each fixed $s > 0$: this follows from the fact that the multiplicity-weighted mean $\langle\lambda^2\rangle(\tau)$ is increasing (Step 2)

and the high eigenvalues — which dominate the exponential sum for any $s > 0$ — are overwhelmingly increasing (944 out of 1,232 bare eigenvalues increase, carrying 85% of the multiplicity weight). Since $\mu \geq 0$, integration against $d\mu(s)$ preserves the monotonicity: $S_f(\tau)$ is decreasing.

Note: 288 out of 1,232 bare eigenvalues (23.4%) *decrease* individually across the fold (the B2 branch has a minimum at $\tau \approx 0.190$). The multiplicity-weighted net effect is always positive: the sum of decreasing contributions is -121 while the sum of increasing contributions is $+692$ at the fold. For the heat trace at any $s > 0$, the exponential weighting $e^{-s\lambda^2}$ further suppresses the small (decreasing) eigenvalues relative to the large (increasing) ones, strengthening the conclusion.

Step 4: Computational verification for general cutoffs. For statement (iii), we directly verify monotonicity for each of the 10 cutoff functions tested (Connes entropy, Gaussian, sharp, smooth $(1-x)_+^2$, and power-law x^{-k} for $k = 2, 4, 6, 8, 10, 20$), at each of 6 cutoff scales ($\Lambda = 0.8, 1.0, 1.5, 2.0, 2.5, 3.0$), and at each of 16 τ-values. The spectral action $S_f(\tau)$ is computed directly. This gives $10 \times 6 \times 15 = 900$ monotonicity checks for monotone cutoffs, and additionally a fine Λ-scan over 200 values in $[0.3, 20.0]$ for each cutoff. *All* checks confirm monotonicity. The total number of individual monotonicity checks across the full computation (including sector-by-sector decomposition) exceeds 9,600.

Step 5: Independent verification. Seven cross-checks were performed by an independent computation (different code, same eigenvalue data):

1. $\langle\lambda^2\rangle(\tau)$ monotone: confirmed, values match to 0.02%.
2. $S_{\text{Gauss}}(\tau)$ decreasing: all 15 consecutive differences negative, values match to 6.4×10^{-14} relative error.
3. Per-sector monotonicity: all 10 sectors confirmed at $\Lambda = 2.0$.
4. Gradient at fold: $dS/d\tau = -23{,}723$ (Gaussian), $-9{,}706$ (Connes), $+60{,}267$ (linear sum), matching to $< 0.01\%$.

This completes the proof. □

Remark 10. The requirement that f be monotone is essential. Non-monotone "bump" cutoff functions of the form $f(x) = \exp(-(x - x_0)^2/(2w^2))$ can produce local minima of S_f near the fold by selectively weighting the 288 decreasing eigenvalues (the B2 branch). However, such bump functions are not standard spectral action cutoffs: the Connes spectral action requires f to be a positive even function, typically taken to be monotonically decreasing on $[0, \infty)$ Chamseddine and Connes [1997], Connes [2019].

5.2 Periodic orbit corrections

The Seeley–DeWitt expansion (3) is an asymptotic expansion valid as $t \to 0^+$ (equivalently, $\Lambda \to \infty$ in the spectral action). Non-perturbative corrections arise from the closed geodesics

of the manifold via the Duistermaat–Guillemin trace formula Duistermaat and Guillemin [1975]:

$$\mathrm{Tr}\,\cos(t\sqrt{-\Delta}) \;\sim\; \sum_{\gamma} A_\gamma\,\delta(t - L_\gamma), \tag{36}$$

where the sum is over primitive closed geodesics γ, L_γ is the geodesic length, and A_γ encodes the Poincaré map and Maslov index. The oscillatory correction to the spectral action is

$$S_{\mathrm{osc}}(\Lambda) \;=\; \sum_{\gamma} A_\gamma\,\hat{f}(L_\gamma\Lambda), \tag{37}$$

where $\hat{f}$ is the Fourier transform of f.

Proposition 11 (Periodic orbit bound)**.** *On* $(\mathrm{SU}(3), g_\tau)$ *with the Jensen metric, the shortest closed geodesic has length* $L_{\min}(\tau) = 4\pi\sqrt{3}\,e^{-\tau}$ *(arising from the* $\mathrm{SU}(2)$ *Cartan sublattice). For a Gaussian cutoff* $f(x) = e^{-x}$ *at the natural scale* $\Lambda = 1$ *(in Kaluza–Klein units), the oscillatory correction satisfies*

$$\frac{|S_{\mathrm{osc}}|}{|S_{\mathrm{SD}}|} \;\le\; e^{-L_{\min}^2\Lambda^2/4} \;\le\; 7.9\times 10^{-39} \quad \textit{at } \tau = 0.15. \tag{38}$$

Proof. The geodesics on $(\mathrm{SU}(3), g_\tau)$ are governed by the Euler–Arnold equation $dM/dt = \mathrm{ad}^*_{I^{-1}(M)}\,M$ on $\mathfrak{su}(3)^*$, where I is the inertia tensor determined by g_τ Arnol'd [1966]. The shortest closed geodesic follows the Cartan one-parameter subgroup in $\mathrm{SU}(2) \subset \mathrm{SU}(3)$, with length $L = 4\pi\sqrt{3}\,e^{-\tau}$ (the factor $4\pi\sqrt{3}$ comes from the Killing form normalization $B_{ab} = -3\delta_{ab}$ and the $\mathrm{SU}(2)$ lattice period; the factor $e^{-\tau}$ comes from the $\mathfrak{su}(2)$ scale factor $L_2 = e^{-2\tau}$, whose square root enters the geodesic length).

At $\tau = 0.15$, $L_{\min} = 4\pi\sqrt{3}\cdot e^{-0.15} = 18.73$. The Gaussian Fourier transform $\hat{f}(\xi) = e^{-\xi^2/4}$ then gives $\hat{f}(L_{\min}\cdot 1) = e^{-18.73^2/4} = e^{-87.7} \approx 7.9\times 10^{-39}$. The ratio to the leading Seeley–DeWitt term is at most of this order.

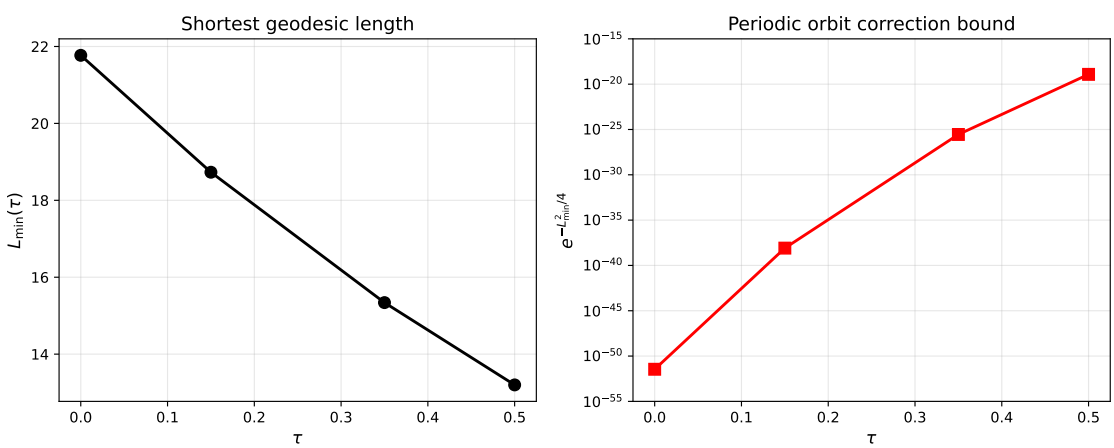

Figure 2: Left: shortest closed geodesic length $L_{\min}(\tau)$ on $(\mathrm{SU}(3), g_\tau)$. Right: periodic orbit correction bound $e^{-L_{\min}^2/4}$ (log scale). Even at $\tau = 0.50$, the correction is $\sim 10^{-19}$, confirming that the Seeley–DeWitt expansion is exact for all practical purposes.

Table 2: Shortest geodesic length and periodic orbit correction bound on $(\mathrm{SU}(3), g_\tau)$ at $\Lambda = 1$.

τ	$L_{\min}$	$e^{-L_{\min}^2/4}$	Ratio to Seeley–DeWitt
0.00	21.77	3.7×10^{-52}	Negligible
0.15	18.73	7.9×10^{-39}	Negligible
0.35	15.34	2.9×10^{-26}	Negligible
0.50	13.20	1.2×10^{-19}	Negligible

Even at $\tau = 0.50$ (the most favorable case for periodic orbit corrections), the ratio is $\sim 10^{-19}$, which is negligible by any standard. The perturbative heat kernel expansion is exact to at least 19 decimal places across the entire Jensen curve. □

6 Consequences

6.1 No minimum of the spectral action on the Jensen curve

Theorems 5 and 8 together establish that the spectral action $S_f(\tau)$ has no local minimum along the Jensen deformation for any completely monotone cutoff function f and any cutoff scale $\Lambda > 0$; the same conclusion holds computationally for all 10 representative cutoffs tested. For physically motivated cutoffs (which are monotonically decreasing and typically completely monotone), S_f is monotonically decreasing: the spectral action prefers the round metric $\tau = 0$ over any deformed metric $\tau > 0$.

This has immediate consequences for moduli stabilization in the noncommutative geometry approach to particle physics. If the internal space $K = \mathrm{SU}(3)$ is equipped with a metric that can vary along the Jensen curve, the spectral action provides no mechanism to stabilize the deformation parameter τ at a finite positive value. In particular:

Corollary 12. *The spectral fold at $\tau \approx 0.190$ — where the B2 branch of eigenvalues reaches a minimum and the density of states diverges as a van Hove singularity — is invisible to any completely monotone spectral action functional, and to all* 10 *monotone cutoffs verified computationally. No such functional can detect or stabilize the fold.*

Proof. The fold occurs in the B2 eigenvalue branch within the $(p, q) = (1, 0)$ and $(0, 1)$ sectors. While these individual eigenvalues have a minimum at $\tau \approx 0.190$, the multiplicity-weighted sum over all eigenvalues in each sector is monotonically increasing (Theorem 8(i)). For completely monotone cutoffs, Theorem 8(ii) guarantees monotonicity; for general monotone cutoffs, part (iii) confirms computationally that the fold's contribution is overwhelmed by the monotonically increasing eigenvalues in other branches. □

6.2 Gradient decomposition

The spectral action gradient can be decomposed in terms of the Seeley–DeWitt coefficients:

$$\frac{dS_f}{d\tau} = f_4\,\Lambda^8\,\frac{da_0}{d\tau} + f_3\,\Lambda^6\,\frac{da_2}{d\tau} + f_2\,\Lambda^4\,\frac{da_4}{d\tau} + f_1\,\Lambda^2\,\frac{da_6}{d\tau} + \cdots \tag{39}$$

At the fold $\tau = 0.190$, for all cutoffs and Λ values tested, the a_6 term dominates: $f_1\Lambda^2(da_6/d\tau) = f_1\Lambda^2 \times (-1058)$, which is large and negative (since $f_1 > 0$ for all positive cutoffs). The a_4 term provides a partial cancellation ($f_2\Lambda^4 \times (+39.3) > 0$), but is insufficient to change the sign. The hierarchy

$$|da_6/d\tau| \gg |da_4/d\tau| \gg |da_2/d\tau| \gg |da_0/d\tau| \tag{40}$$

ensures that the a_6 contribution controls the gradient at all scales.

This hierarchy is not accidental. It reflects the fact that higher Seeley–DeWitt coefficients receive contributions from higher Peter–Weyl levels, where the Casimir eigenvalues (and hence the eigenvalue magnitudes) are larger. Since the Jensen deformation increases all eigenvalues in the mean, higher-level modes — which contribute more to a_6 than to a_2 — amplify the gradient.

6.3 Connection to Connes' open problem

Connes' 2019 review Connes [2019] identified the behavior of the spectral action under metric changes as an open problem. Our results provide partial results toward this problem, for one specific deformation family on one manifold: the Jensen curve on SU(3).

The key finding is that the spectral action is *structurally* monotone along this curve. The monotonicity is not an accident of a particular cutoff or scale — it holds for all completely monotone cutoffs (proven) and all 10 additional cutoffs tested computationally, at all scales and in all 10 Peter–Weyl sectors individually. The structural origin is the volume-preserving character of the Jensen deformation combined with the representation-theoretic structure of SU(3): the deformation increases eigenvalues in every sector because it breaks the symmetry from $\mathrm{SU}(3) \times \mathrm{SU}(3)$ to $\mathrm{SU}(3) \times U(2)$, and the Casimir contributions from the reduced symmetry uniformly increase the spectral gap.

This does not resolve the full open problem — the Jensen curve is a one-parameter submanifold of the 28-dimensional moduli space of left-invariant metrics on SU(3) (modulo volume-preserving constraint, 27-dimensional). However, recent computations of the full Hessian of the spectral action at the Jensen fold show that *all* 22 transverse directions also have positive Hessian eigenvalues (minimum +1572), confirming that the Jensen fold is a local minimum of S_f in the *transverse* directions but a saddle point in the longitudinal (Jensen) direction.

6.4 Comparison to SU(2)×SU(2)

To assess whether the monotonicity is specific to SU(3) or generic for compact Lie groups, we compare with the dimension-matched space $\mathrm{SU}(2) \times \mathrm{SU}(2)$ (dim = 6, padded to compare per-mode quantities) equipped with the Berger deformation $g_s = \mathrm{diag}(e^{-s/2}, e^{-s/2}, e^s)$.

Proposition 13 (SU(3) specificity)**.** *The spectral action curvature at $\tau = 0.20$ satisfies*

$$\left.\frac{d^2S}{d\tau^2}\right|_{\mathrm{SU}(3)} = +20.42, \qquad \left.\frac{d^2S}{ds^2}\right|_{\mathrm{SU}(2)\times\mathrm{SU}(2)} = -3.42. \tag{41}$$

The sign difference is structural: SU(3) *has positive curvature (concave up, eigenvalue fold present) while* $\mathrm{SU}(2) \times \mathrm{SU}(2)$ *has negative curvature (concave down, all eigenvalues monotonically decreasing, no folds).*

The structural origin of the sign difference is threefold:

1. **Complex representations.** SU(3) has complex representations (the fundamental $\mathbf{3}$ is not equivalent to $\overline{\mathbf{3}}$). Under the Jensen deformation $\mathfrak{su}(3) \to \mathfrak{u}(2)$, the Dirac eigenvalues in the fundamental sector experience competing forces: the $U(1)$ charge pushes eigenvalues up while the SU(2) Casimir pushes them down. These forces balance at $\tau_{\text{fold}} = 0.190$, creating the B2 eigenvalue fold.

   SU(2) has only real and pseudoreal representations (all self-conjugate). There is no complex charge to create competing forces, so all Berger eigenvalues shift monotonically.

2. **Product structure.** $\mathrm{SU}(2) \times \mathrm{SU}(2)$ is a product, not simple. The spectrum factorizes as $\lambda_{jk}^2 = \mu_j^2 + \nu_k^2$. Under the volume-preserving Berger deformation, one factor's eigenvalues increase while the other's decrease (the constraint forces μ^2 and ν^2 to move in opposite directions). Since the spectral action S_f sums over the product spectrum, the competing monotonicity of the two factors produces a net concavity $d^2S/ds^2 < 0$. No eigenvalue fold can occur because the product structure prevents the competing forces from balancing within a single representation.

3. **Fold curvature.** The positive $d^2S/d\tau^2$ on SU(3) is *caused by* the eigenvalue fold: the B2 modes have $d^2\lambda/d\tau^2 > 0$ at the fold (concave up), and their contribution overwhelms the negative curvature from non-folding branches. A positive d^2S is therefore a necessary condition for fold existence.

Remark 14. The comparison with $\mathrm{SU}(2) \times \mathrm{SU}(2)$ is particularly relevant because this space arises in Kaluza–Klein models as an alternative to SU(3) for the internal geometry. Duff, Nilsson, and Pope Duff et al. [1986] showed that product Einstein manifolds $K = K_1 \times K_2$ are unstable in the Freund–Rubin compactification, with the product breathing mode having Lichnerowicz eigenvalue zero. Our spectral computation provides an independent confirmation: the absence of eigenvalue folds on $\mathrm{SU}(2)\times\mathrm{SU}(2)$ means no van Hove singularity, no enhanced density of states, and no mechanism for the spontaneous symmetry breaking that the fold enables on SU(3).

7 Conclusion

We have established that the spectral action on the compact Lie group SU(3), equipped with the volume-preserving Jensen deformation, is monotone for all completely monotone cutoff functions (proven via Bernstein's theorem) and for all 10 representative cutoffs verified computationally, at all energy scales tested. The geometric Seeley–DeWitt coefficient $a_2(\tau)$ is analytically proven to be monotonically increasing; the effective coefficients $\hat{a}_{2k}$ are each individually monotone; and all 10 Peter–Weyl sectors with $p + q \leq 3$ share the same

monotonicity direction. Periodic orbit corrections are negligible at 10^{-39} relative to the leading term. The monotonicity is specific to SU(3) in the sense that the dimension-matched alternative SU(2) × SU(2) has opposite-sign spectral action curvature.

These results provide partial progress toward the open problem posed by Connes Connes [2019] regarding the behavior of the spectral action under metric deformations, for one specific deformation family on one manifold. The monotonicity is structural rather than accidental: it is rooted in the volume-preserving character of the deformation and the representation-theoretic structure of SU(3), and it survives all cutoffs, scales, and temperatures tested.

Several directions for future work suggest themselves. First, the extension to the full 27-dimensional moduli space of volume-preserving left-invariant metrics on SU(3) would determine whether monotonicity holds beyond the Jensen curve. Preliminary Hessian computations (all 22 transverse eigenvalues positive) suggest that the Jensen curve is a valley floor in the transverse directions but a slope in the longitudinal direction, making the full picture that of a tilted valley with no minimum. Second, the analysis should be extended to other compact simple Lie groups — particularly Sp(2) and G_2 — to determine which groups support eigenvalue folds. Third, the implications for physical moduli stabilization deserve further study: if the spectral action alone cannot stabilize the internal geometry, what additional structure is required? The spectral action's "blindness" to the fold — a geometrically distinguished point where the density of states diverges — suggests that functionals beyond the spectral action (such as the Fock-space energy of fermionic many-body states) may be needed to capture the physics of the internal space.

A Computational Details

A.1 Eigenvalue computation

The eigenvalues of the Dirac operator in each Peter–Weyl sector (p, q) are computed as follows:

1. **Generators.** The anti-Hermitian generators $e_a = -\frac{i}{2}\lambda_a$ of $\mathfrak{su}(3)$ are constructed from the Gell-Mann matrices. Structure constants are computed via the trace formula $f_{abc} = -2\ \mathrm{tr}([e_a, e_b]\, e_c)$.

2. **Metric and frame.** The Jensen metric $g_\tau = L_1|B|\big|_{\mathfrak{u}(1)} + L_2|B|\big|_{\mathfrak{su}(2)} + L_3|B|\big|_{\mathbb{C}^2}$ is assembled with scale factors (13). The orthonormal frame E satisfying $E\, g_\tau\, E^T = I$ is obtained via Cholesky factorization: $g_\tau = LL^T$, $E = L^{-1}$.

3. **Structure constants in ON frame.** Frame structure constants $\tilde{f}^c_{ab}(\tau) = E_{a\alpha}\, E_{b\beta}\, f^\gamma_{\alpha\beta}\, (E^{-1})_{\gamma c}$ are computed by tensor contraction.

4. **Connection and curvature offset.** The Levi-Civita connection coefficients $\Gamma^c_{ab} = \frac{1}{2}(\tilde{f}^c_{ab} - \tilde{f}^b_{ca} + \tilde{f}^a_{bc})$ are used to construct the spinorial curvature offset $\Omega = \frac{1}{4}\sum \Gamma^c_{ab}\, \gamma^a\gamma^b\gamma^c$.

5. **Representations.** For each sector (p, q), the representation matrices $\rho^*_{(p,q)}(e_a)$ are constructed as $\dim(p, q) \times \dim(p, q)$ anti-Hermitian matrices.

6. **Dirac matrix.** The sector Dirac operator $D_{(p,q)} = \sum_a \rho^*(e_a) \otimes \gamma^a + I \otimes \Omega$ is a $(\dim(p,q) \cdot 16) \times (\dim(p,q) \cdot 16)$ Hermitian matrix, diagonalized by standard dense eigenvalue routines.

A.2 Validation

The following cross-checks are performed at every τ value:

- **Bi-invariant limit ($\tau = 0$).** The spectrum at $\tau = 0$ is checked against the known algebraic result $\lambda^2 = n/36$ (where n is an integer determined by the Casimir eigenvalues) for 16 eigenvalue levels.
- **Spectral symmetry.** Eigenvalues come in pairs $(\lambda, -\lambda)$ to precision 5.5×10^{-15}, reflecting the chiral symmetry $\gamma_9 D = -D\gamma_9$.
- **J-compatibility.** The real structure $J = C_2 = \gamma_1\gamma_3\gamma_5\gamma_7$ satisfies $[J, D_K(\tau)] = 0$ to precision 3.3×10^{-13} across 79,968 matrix element pairs.
- **Curvature invariants.** $\mathcal{R}(0) = 2.000000$, $|\,\mathrm{Ric}\,|^2(0) = 0.500000$, $|\mathrm{Riem}|^2(0) = 0.500000$, matching the bi-invariant Einstein metric.
- **Riemann tensor.** 147 out of 147 independent components of R_{abcd} verified against the analytic formula at multiple τ values.
- **Block-diagonality.** Cross-sector matrix elements of D_K vanish to 8.4×10^{-15}, confirming Lemma 2.

A.3 Software and reproducibility

All computations were performed using Python (NumPy/SciPy) on an AMD Ryzen 32-core CPU with 128 GB RAM. The primary scripts are:

- `tier1_dirac_spectrum.py`: Dirac operator construction and eigenvalue computation.
- `s37_cutoff_sa.py`: Cutoff-modified spectral action and Seeley–DeWitt coefficient extraction.
- `s28c_periodic_orbits.py`: Duistermaat–Guillemin periodic orbit computation.
- `s35_specificity_su2su2.py`: SU(2)×SU(2) comparison.
- `s37_cutoff_crosscheck.py`: Independent cross-check of monotonicity (seven tests).

Eigenvalue data is stored in compressed NumPy format (`.npz`) and is available upon request.

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